How-To Library

Informatica

# Running a Column Profile on JSON and XML Data Sources

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# Abstract

When you run a profile on JSON and XML data sources, Informatica Developer treats the files as flat files. This article describes how you can create column profiles on flat file data sources that are in JSON or XML format, and view the profile results for data analysis and development.

# Supported Versions

- Profiling and Discovery 10.0

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# Overview

You can create column profiles on flat file data sources that use the CSV format or on semi-structured data sources that use the JSON or XML format.

JSON and XML formats are widely used in databases and in data interchange. JSON and XML use a hierarchical format to transmit structured data over a network connection.

A profile finds and analyzes all data irregularities across data sources in the enterprise and hidden data problems that put data projects at risk. Running a profile on any data source in the enterprise gives you a good understanding of the strengths and weaknesses of its data and metadata.

To detect data irregularities in a data source that uses JSON or XML format, perform the following tasks:

1. Create a flat file data object or complex file data object on the data source.
2. Create a column profile on the data object.

**Note:** You can create, edit, and view the column profile for JSON or XML data sources in the Developer tool. You can view the profile in the Analyst tool in summary view and detailed view.

# Types of Data Objects

You can create a flat file data object or complex file data object for data sources in JSON or XML format.

You can create a complex file data object with access type as file or connection. You can choose flat files as data sources when the access type is a file. You can choose JSON or XML data sources in Hadoop Distributed File System (HDFS) when the access type is connection.

You can create the following types of data objects for data sources in JSON or XML format:

- Flat file data object.
- Complex file data object for flat files.
- Complex file data object for JSON and XML data sources in HDFS.
- Complex file data object for multiple JSON or XML files in HDFS that are saved in a folder.

# Flat File Data Object

Informatica Developer cannot read the flat file with JSON or XML format directly. You must create a text file, add the data source location into the text file, and create a flat file data object with the text file.

You can add file locations for multiple JSON or XML data sources in the text file. However, you must add the location for those files that are in the same format. A run-time error occurs when the text file contains file locations for files that are in different formats.

## *Creating a Flat File Data Object With a Text File*

You can create a flat file data object with a text file that contains the location of the source JSON or XML file.

1. In the **Object Explorer** view in the Developer tool, select the project where you want to create the data object and column profile.
2. Click **File** > **New** > **Data Object**.

   The **New** dialog box appears.
3. Select **Physical Data Objects** > **Flat File Data Object**, and click **Next**.

   The **New Flat File Data Object** dialog box appears.
4. Select **Create from an Existing Flat File**, and click **Browse** to choose the text file. Click **Next**.
5. Verify that the code page is **MS Windows Latin 1 (ANSI), superset of Latin 1**, and the format is delimited. Click **Next**.
6. Verify that the delimiter is set to **comma**. Click **Finish**.

# Complex File Data Object for Flat Files Overview

Informatica Developer can read data sources with JSON or XML format through a complex file reader. The complex file reader provides input to a Data Processor transformation that parses the file and converts the source data to flat CSV

records. Data Processor transformation is used to process messaging formats, HTML, XML, JSON, and other similar formats.

You can use a complex file data object for the following types of data sources:

- Flat files that use JSON or XML formats.
- Files that use JSON or XML formats and are stored in HDFS.

### *Creating a Complex File Data Object with a JSON or XML File*

You can create a complex physical data object for the JSON or XML source file, and then create a column profile on the physical object.

1. In the **Object Explorer** view, select the project.
2. Click **File** > **New** > **Data Object**.

   The **New** dialog box appears.
3. Select **Physical Data Objects** > **Complex File Data Object**, and click **Next**.

   The **New Complex File Data Object** dialog box appears.
4. Add a name for the data object. Select the access type as **File**.
5. Click **Browse** to choose a JSON or XML file. Click **Finish**.

   When the Developer server is in Linux, you must update the file path of the data source to the location in the server. To update the file path, select the complex file data object, click **Read** in the **Data Object Operations** tab, and add the file path in the **Advanced** tab in the **Data Object Operation Details** pane.

   The data object appears in the project folder.

## Complex File Data Object For Data Sources In HDFS

To create a complex file data object for data sources in HDFS, you must create a connection for Informatica Developer to accept files in HDFS. You can then create a complex file data object using this connection.

### *Creating a HDFS Connection*

To create a column profile on a JSON or XML source in HDFS, the Informatica Developer must accept HDFS connections. You can create a complex file data object after you create a connection with HDFS.

1. Click **Window** > **Preferences** > **Informatica** > **Connections** > **File Systems** > **Hadoop File System**. Select a HDFS connection.

   The **Edit Connection** dialog box appears.
2. In the **Edit Connection** dialog box, enter the **NameNode URI**. Click **OK**.
3. In the **Preferences** dialog box, click **OK**.

### *Creating a Complex File Data Object with a JSON or XML File in HDFS*

You can create a complex physical data object for the JSON or XML source file that uses HDFS and create a column profile on the physical object.

1. In the **Object Explorer** view in the Developer tool, select the project where you want to create a physical data object and column profile.
2. Click **File** > **New** > **Data Object**.

   The **New** dialog box appears.

3. Select **Physical Data Objects** > **Complex File Data Object**, and click **Next**.

   The **New Complex File Data Object** dialog box appears.
4. Add a name for the data object. Select the access type as **Connection** .
5. Click **Browse** to select a connection. In the **Add Resource** dialog box, click **Add** to choose a JSON or XML file. Click **Finish**.

   The data object appears in the project folder.

# Complex File Data Object with JSON or XML Files in a Folder

You can create a folder and place all the data sources that use JSON or XML format and in HDFS. You can then create a complex file data object with the folder. You can use this scenario when the data source file size is more than 1GB. You can split the file into multiple files and place them in the same folder and create a data object on the folder. If the file size is equal to or greater than 2GB, then configure the Maximum Heap Size option to 2GB for Data Integration Service.

## *Creating a Complex File Data Object with JSON or XML Files in a Folder*

You can create a column profile on a folder that has multiple JSON or XML files in HDFS.

1. In the **Object Explorer** view, select the project where you want to create a physical data object and column profile.
2. Click **File** > **New** > **Data Object**.

   The **New** dialog box appears.
3. Select **Physical Data Objects** > **Complex File Data Object**, and click **Next**.

   The **New Complex File Data Object** dialog box appears.
4. Add a name for the data object. Select the access type as **Connection** .
5. Click **Browse** to select a connection. In the **Add Resource** dialog box, click **Add** to choose a JSON or XML file in the folder. Click **Finish**.

   The data object appears in the project folder.
6. Click **Advanced** > **Runtime: Read** > **Source file directory**. Remove the file name and retain the folder name in the file path.

# Running a Column Profile on JSON or XML Data Sources

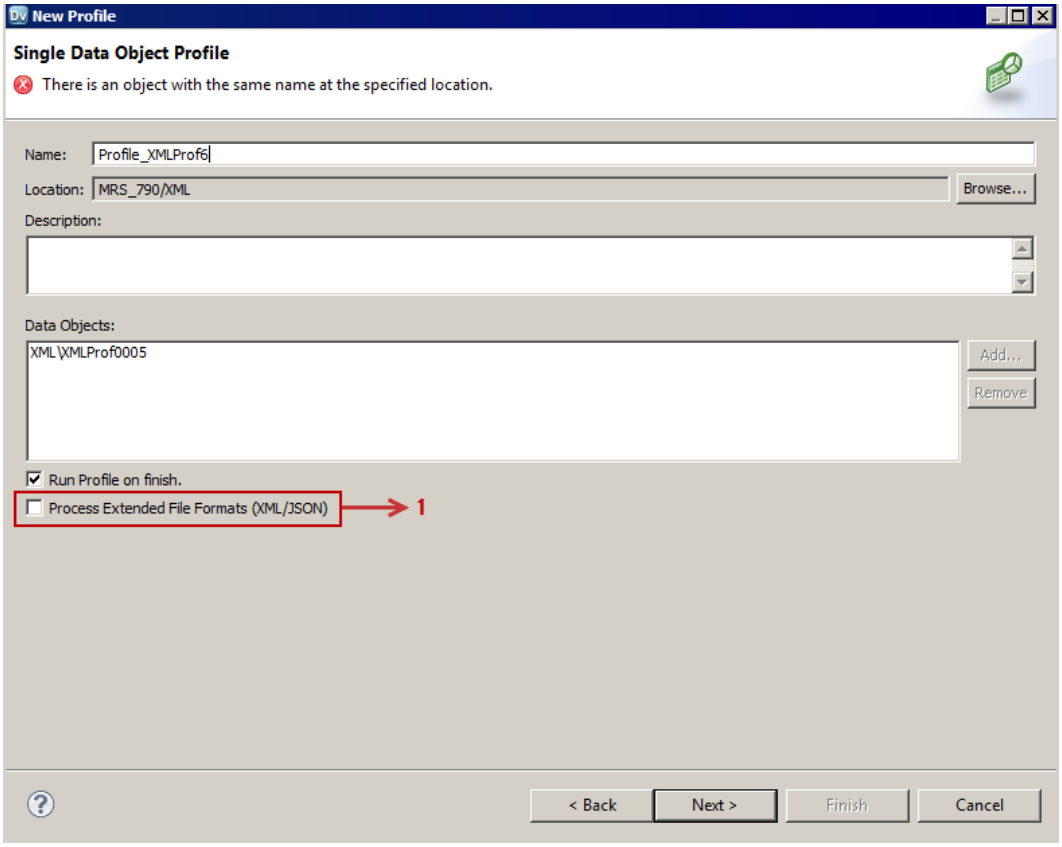
After you create a flat file data object or complex file data object with JSON or XML data sources, you can create a column profile on the data object.

1. In the Object Explorer view, select the physical data object for the JSON or XML file.
2. Click **File** > **New** > **Profile**.

   The **New** dialog box appears.
3. Select **Profile**. Click **Next**.

   The **New Profile** dialog appears.
4. In the **New Profile** dialog box, add a name for the profile and an optional description.
5. Select **Process Extended File Formats (XML/JSON)** option. Click **Next**.

The following image displays the **New Profile** wizard with the **Process Extended File Formats (XML/JSON)** option:




1. Process Extended File Formats (XML/JSON). Select this option to process data sources in JSON or XML format.

6. In the **Single Data Object Profile** page, select the columns and options under **Column Selection** and **Data Domain Discovery** as required. Click **Finish**.

   **Note:** When the Developer server is in Linux and the physical data object is a flat file data object with a text file, then you must:

   1. Update the **Precision** value in the **Overview** tab to include the number of characters in the file path of the data source in the server.
   2. Update the file path of the data source to the location in the server after you create a profile on the flat file data object. To update the file path, click **Runtime: Read** > **Source file directory** in the **Advanced** tab, and add the file path.

7. Right-click the profile, and select **Run Profile**.

   The profile results appear.

# JSON or XML Column Profile Results Overview

You can view profile results in the Analyst tool or the Developer tool.

## *JSON and XML Profile Results on the Analyst Tool*

In the Analyst tool, you can view the profile results in the summary view and you can view column-specific information in the detailed view.

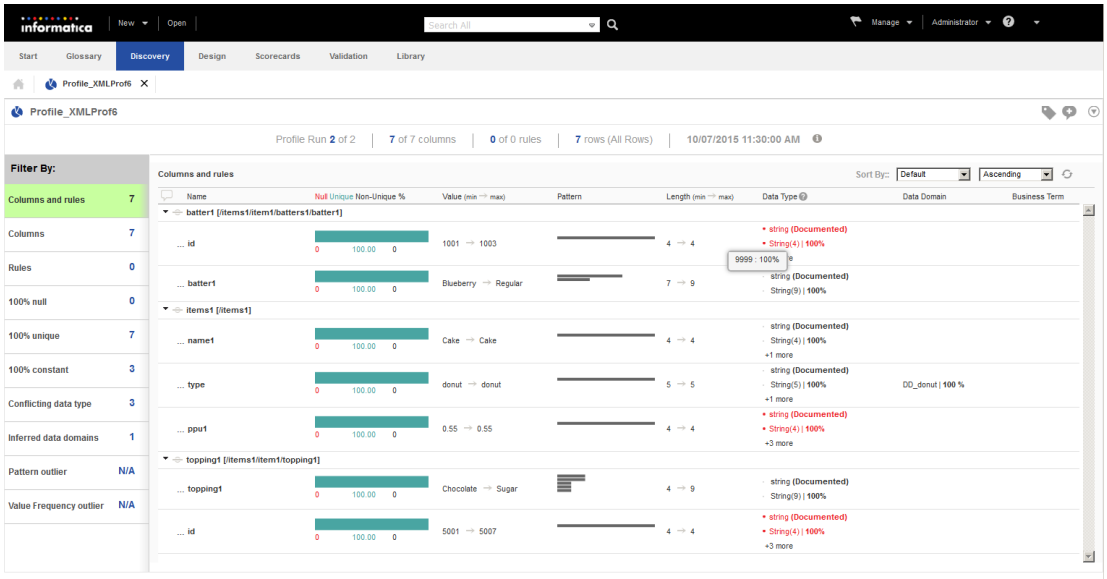
### Summary View of JSON and XML Profile Results

In the summary view, you can view the profile results by using the default filters.

The following parameters are displayed as horizontal bar charts and in percentages:

- Name
- Null, unique, non-unique values
- Pattern
- Minimum and maximum values
- Minimum and maximum length of the values
- Data type
- Data domain
- Business term

The following image displays the summary view of a JSON profile:




You can run the profile multiple times and compare two profile runs. You can view the profile results for a historical profile run and view the latest column information through the incremental profile run. You can add comments to a profile or to a column. You can export the profile results to a Microsoft Excel file.

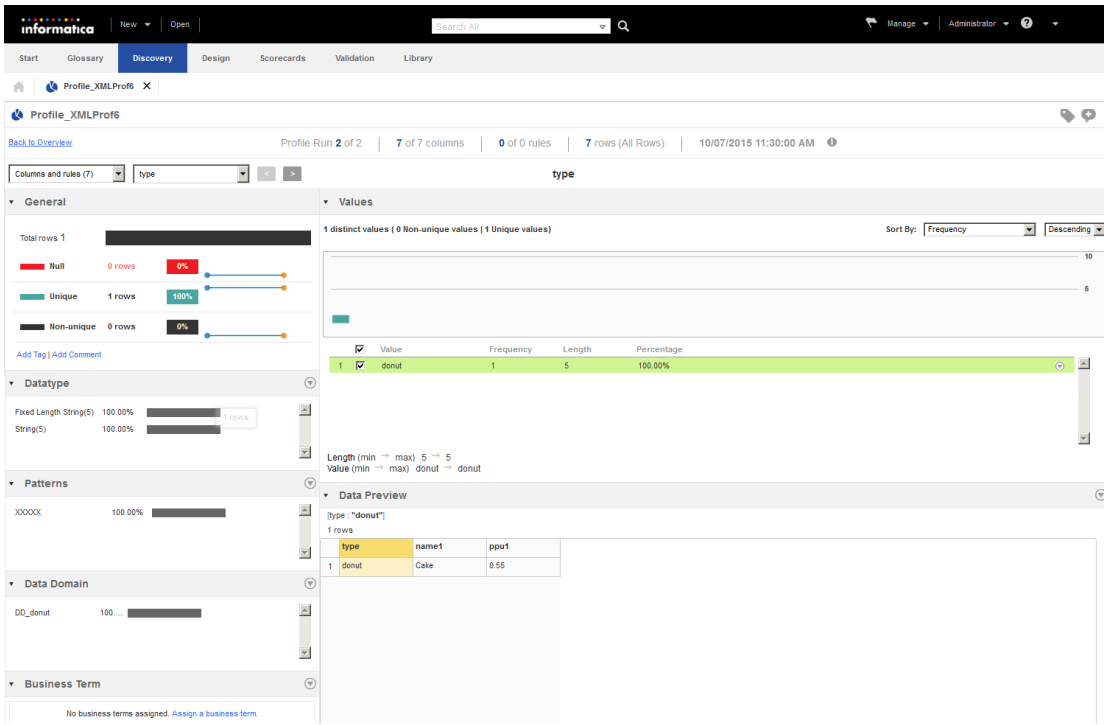
## Detailed View of JSON and XML Profile Results

In the detailed view, you can view column-specific results.

You can perform the following tasks in detailed view:

- View sparkline. The graph displays the increase and decrease of null, unique, and non-unique values in 5 consecutive profile runs.
- View inferred data types and data domains.
- Curate data types and data domains.
- View values in detail in the **Values** pane.
- Drill down on data.
- Assign business terms.

The following image displays the detailed view of a column in a JSON profile:

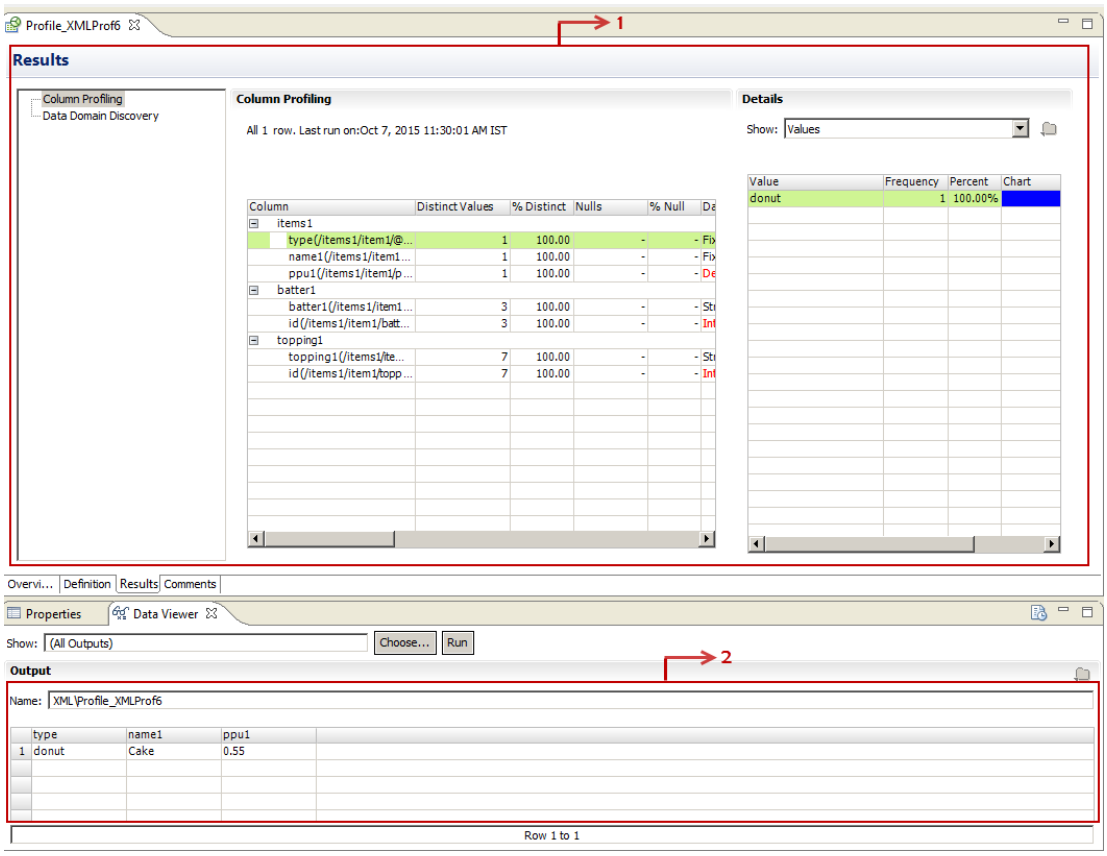



## *JSON and XML Profile Results on the Developer Tool*

In the Developer tool, you can view the results in the Results pane.

You can drill down on the results, add comments, and create scorecards for the profile.

The following image displays the profile results in the Results pane and the drilldown results in the Output pane:




1. Results pane. You can view the profile results in the Results pane.
2. Output pane. You can view the drilldown results in the Output pane.

# Authors

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